

PUBLIC-PROJECT MECHANISM DESIGN • CERTIFICATE-FIRST RESEARCH

When the “fair” funding rule isn’t

An exact map of how small groups can break the textbook rule for funding shared projects — and by exactly how much.

Mohit Mendiratta • Independent research • SSRN Abstract ID 7293498

github.com/jammysunshine/when-fair-funding-isnt

How do you fairly split the bill for something everyone benefits from?

- Picture an HOA board deciding whether to fund a new roof, a neighborhood group crowdfunding a footbridge, or a DAO voting to spend from a shared treasury.
- The standard textbook rule (the “pivotal mechanism”) says: build the project if pledges cover the cost, and charge each person only the smallest amount that was actually necessary for their vote to matter.
- This rule is provably honest for individuals — nobody benefits from lying about what they value the project at, acting alone.
- This paper asks the next question nobody had answered with an exact formula: what happens when a small group coordinates their pledges together?

Groups can game the textbook rule — this paper proves exactly when

- The “fair” critical-payment rule is not fair against coordinated groups: in the tested cases, coalitions of just two or three people broke it in the large majority of configurations.
- This paper proves — not just tests — exactly when that happens: whenever the project's cost is low enough that any one member's pledge is dispensable, the whole group can jointly game the price to zero.
- A simple, checkable formula tells a designer, before launch, whether their specific group size, pledge cap, and project cost are exposed — and exactly how much a colluding group stands to gain.
- Every claim is backed by an executable, independently re-checked proof — not a simulation and not an opinion.

Three neighbors, a \$3,000 footbridge

Honestly: the three neighbors truly value the bridge at \$0, \$1,000 and \$1,000 — pledges totaling only \$2,000. That's short of the \$3,000 cost, so nothing gets built and nobody pays a cent.

Colluding: all three instead agree to each claim \$1,500, regardless of their true value. Reported pledges now total $\$4,500 \geq \$3,000$, so the bridge gets built.

The bill: the textbook rule charges each neighbor only what was needed to make their vote pivotal. Drop any one neighbor's \$1,500 claim, and the other two still claim $\$1,500 + \$1,500 = \$3,000$ — already enough on its own. So every neighbor's charge is $\$3,000 - \$3,000 = \$0$.

- Result: a \$3,000 bridge gets built, and the group pays \$0 total — the mechanism runs a \$3,000 deficit.
- This isn't special to three people or \$1,500 — it happens whenever a colluding group's own inflated claims alone already clear the cost.
- This is not a bug in one bad rule — it is the standard, textbook “pivotal mechanism” taught in every mechanism-design course.

Exhaustive, machine-checked proof — not a sampled experiment

- **Exact enumeration.** Every possible allocation rule in the studied finite class is checked — all of them, not a sample — against six formal fairness and honesty properties.
- **A proven theorem.** A closed-form rule (not just a table of results) predicts exactly which allocation rules survive, for any group size and any pledge cap, and is confirmed against a larger untouched test domain.
- **Independent re-verification.** A second, separately written program re-derives every result from the raw data without reusing any of the first program's code — catching the kind of error a single implementation could hide.
- **An exact formula for group manipulation.** A closed-form expression for exactly how much a coalition of a given size can gain, derived by mathematical proof (convexity), not search — and it matches 145 independently brute-force-checked cases exactly.
- **A designer's checklist.** The paper ends with a plain checklist: plug in your group size, pledge cap, and cost, and know before launch whether your specific setup is exposed.

What this paper is, and deliberately is not

IS

- An exact, fully-enumerated finite theorem
- A reproducible, independently re-checked certificate
- A precise formula for coalition risk on this specific rule
- A practical pre-launch checklist for real designers

IS NOT

- A new universal impossibility theorem
- A claim about continuous (non-integer) values
- A repaired, collusion-proof mechanism
- Evidence about any specific deployed platform

Real systems already use this exact rule

- Participatory-budgeting platforms deciding which community proposal to fund.
- HOA and condo boards voting on a shared capital assessment.
- Crowdfunded neighborhood infrastructure — playgrounds, footbridges, shared solar installs.
- DAO treasuries and quadratic-funding platforms that have piloted pivotal-style payment rules.
- Anywhere a small, well-acquainted group decides together whether to build something and splits the bill by “what you were pivotal for” — this is the exact rule in question, and the paper's formula tells you if your setup is exposed.

Methodology and rigor

- **Certificate-first discipline.** Every accepted or rejected rule ships with a machine-checkable witness; nothing is asserted from sampled performance.
- **Independent replay.** A second implementation (`public_project_independent.py`) reconstructs every table and payment from serialized JSON without importing the primary verifier — zero mismatches across 145+ cross-checked rows.
- **SMT cross-checks.** For the neural-network/ReLU audit lane, an external exact-real Z3 solver independently confirms all reported extrema — 18 strict-bound counterexample queries, all unsat.
- **Preregistration.** The finite-lattice theorem and its confirmation domain were committed before the larger $n=3$, $m=4$ run; the coalition and false-name supplements are explicitly labeled post-hoc.
- **Full reproducibility.** Every script, config, and SHA-256 digest is published; the entire study reruns from a single command list.

READ THE FULL PAPER

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Full code, data, and independent verification scripts:

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